

# Tree Shape

Binary Trees, Height, Leaf & Internal Counts · Mentor-Led Lab (Student Edition)

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**Difficulty** ★★ Intermediate

## Problem

Measure a tree's shape: its height in edges, how many leaves it has, and how many internal (non-leaf) nodes.

Implement all three recursive measures from scratch.

## What to Compute

- Height: the longest root-to-leaf path in edges (empty = -1, single node = 0).
- Leaves: nodes with no children.
- Internal: nodes with at least one child.

## Input / Output Format

**Input:** One line: the tree in level order (values and N).

**Output:** Three lines: Height, Leaves, Internal.

## Examples

### Example 1

**Input:** tree = 1 2 3 4 5 N 6

**Output:**

Height: 2  
Leaves: 3  
Internal: 3

**Explanation:** Three full levels give height 2; nodes 4, 5, 6 are leaves and 1, 2, 3 are internal.

### Example 2

**Input:** tree = 1

**Output:**

Height: 0  
Leaves: 1  
Internal: 0

**Explanation:** A single node is height 0, a leaf, and has no internal nodes.

## Constraints

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- Height is in edges (empty = -1, single node = 0).
- A leaf has no children; an internal node has at least one.
- All three are computed recursively.

## Sample Run

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### Sample Input

```
1 2 3 4 5 N 6
```

### Sample Output

```
Height: 2  
Leaves: 3  
Internal: 3
```